Assignment 6 Solution

Q.1. What is a parameterized T-norm also known as?

(a) Parameterized Z-norms

(b) Parameterized S-conorms

(c) Parameterized F-norms

(d) Parameterized G-conorms

(e) None of the above

Answer: (b)

Q.2. In Dombi's class of T-norm, the range of the parameter lambda is:

(a) (0, 1)

(b) (1, infinity)

(c) (-1, 1)

(d) (0, infinity)

(e) None of the above

Answer: (d)

Q.3. What is the range of the parameter alpha in Dubois Prade's class of T-norm?

(a) [0, infinity]

(b) [-1, 1]

(c) [0, 1]

(d) [1, infinity]

(e) None of the above

Answer: (c)

Q.4. In Yager's class of T-norm, how does increasing the parameter w affect the resultant fuzzy set obtained from the intersection of two fuzzy sets?

(a) It decreases the membership values

(b) It has no effect on the membership values

(c) It increases the membership values

(d) It randomizes the membership values

(e) None of the above

Answer: (c)

Q.5. How does increasing the parameter alpha affect the resulting fuzzy set in Dubois-Prade's class of S-norm?

(a) Increases membership values

(b) No significant change in membership values

(c) Decreases membership values

(d) None of the above

Answer: (b)

Q.6. What parameter determines the behaviour of the resulting fuzzy set in Dubois-Prade's class of S-norm?

(a) Alpha

(b) Lambda

(c) w

(d) Theta

(e) None of the above

Answer: (a)

Q.7. What is the Cartesian product of two arbitrary crisp sets (A,B) with universe of discourse X and Y?

(a) A union B

(b) A intersection B

(c) A cross B

(d) A complement B

(e) None of the above

Answer: (c)

Q.8. In a Cartesian product of two crisp sets A and B, how many ordered pairs are obtained if set A has 4 elements and set B has 3 elements?

(a) 6

(b) 7

(c) 10

(d) 12

(e) None of the above

Answer: (d)

Q.9. How is approachability between elements of different sets represented in fuzzy relations?

(a) Using Boolean logic

(b) Through crisp relations

(c) Using membership degrees

(d) By defining clear boundaries

(e) None of the above

Solution: (c)

Q.10. For the intersection of two crisp relations, membership values are determined by:

(a) Taking the average

(b) Taking the minimum

(c) Applying Boolean logic

(d) Using the intersection of sets

(e) None of the above

Solution: (b)